

HOSPITAL FLOW ANALYSIS

BY:
TEAM 3

This analysis provides a clear view into how patients move through the hospital system, highlighting where pressure builds, when congestion occurs, and what this means for operational decision-making.

The main source of pressure is the imbalance between high, unpredictable admissions and relatively fixed discharge timelines.

Over 55% of admissions are emergency cases, meaning patient inflow is largely unplanned and difficult to control. At the same time, the average length of stay remains consistent at approximately 4 days across departments and patient groups. This creates a situation where: Patients continue to arrive unpredictably, but beds are not freed up quickly enough to match demand.

Additionally, a significant number of patients require rehabilitation or external care after discharge, which can delay bed turnover when placements are not immediately available.

Furthermore, Congestion appears to follow predictable time-based and weekly patterns with peak admission hour around 5 AM, and additional surges at around 10 AM and 8 PM. Busiest days are Sundays and Mondays, with consistently high weekend demand.

Monthly, Admission load fluctuates, but demand remains persistently high throughout.

These patterns indicate that while admissions are steady overall, there are clear “rush periods” where demand exceeds immediate capacity.

Also, while demand is relatively evenly distributed across departments, a few departments such as Orthopedics and Pediatrics handle the highest number of patients, and because length of stay is similar across departments, higher volume directly translates to higher pressure in these units.

In summary, the central challenge is not simply “too many patients,” but rather a flow management issue across the entire care pathway. So, to improve performance, leadership should focus on:

- Strengthening early-day staffing and intake processes to handle peak admission hours
- Improving discharge coordination, especially with rehabilitation and external care facilities.
- Enhancing bed management systems to increase turnover responsiveness during surge periods
- Balancing elective scheduling to maintain flexibility when emergency demand spikes.